

backend/rag/vectorstore.py Documentation (Roman Urdu)

Yu script SoloLLM ki "Sub sy Taiz Memory (Bheja / Database)" hs jab hum ne FAISS library lhaai hy.

File Ka Maqsad (Purpose)

Jab hum laakhon chunks ki list rakhte byn tb simple Python Array scanning s search mn minatlgte mn! **Meta's FAISS** library C++ p bani ha or issy million vectors 1 millisecond ms dhunda jay sjts hz. Yeh VectorStore isee FAISS or SQLite metadata ko link map mntain krt ahy!

Code Blocks aur Functions ki Tafseel

1. SQLite Backing (`VECTOR_SCHEMA`)

- **Tafseel:** `document_chunks` r `chunk_vectors` table mn `vector_id` save hn. Jab FAISS similar Vector ID dhoundta hb, tu fr SQLite s index dekh lr uss Vector Ka text/content fetch kia ht jate r.

2. `add_chunks()`

- **Kya karta hai?:** FAISS m index create ar store mart hy arr Numpy arrays (Fast Maths Vectors) dalti. Aur in sbko `L2 Normalize` bi karatt bb!

3. `search()`

- **Kya karta hai?:** `index.search` callrhy aay FAISS k upper, sath user query float. Bohut fast top-K similar cosine results return hotye k!

4. `_migrate_legacy_embeddings()`

- Ybh ek upgrade script f. Agra SoloLLM ki purani versions main vector blob men thay tu usay fast faiss (`.faiss` file) par shift krvtii eh bgair data los.